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3. *(a)* Crude oil can be separated into simpler mixtures called fractions. These fractions contain hydrocarbon compounds called alkanes. **Table 1** shows information about some of the fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C_1-C_4
petrol	30-75	C_5-C_{10}
naphtha	70-170	C_8-C_{12}
kerosene	170-250	C_{10}-C_{14}
diesel oil	250-340	C_{14}-C_{24}
lubricating oil	340-500	C_{21}-C_{30}
fuel oil	490-580	C_{25}-C_{35}
residue	> 580	$>C_{35}$

Table 1

Use **only** the information in **Table 1** to answer parts (i)-(iii).

(i) Hexane has a boiling point of 68 °C. Give the name of the fraction which contains hexane. [1]

....

(ii) One alkane is found in kerosene and in diesel. Give the number of carbon atoms in this alkane. [1]

....

(iii) Give the number of carbon atoms in the alkane which has the **lowest** boiling point. [1]

....

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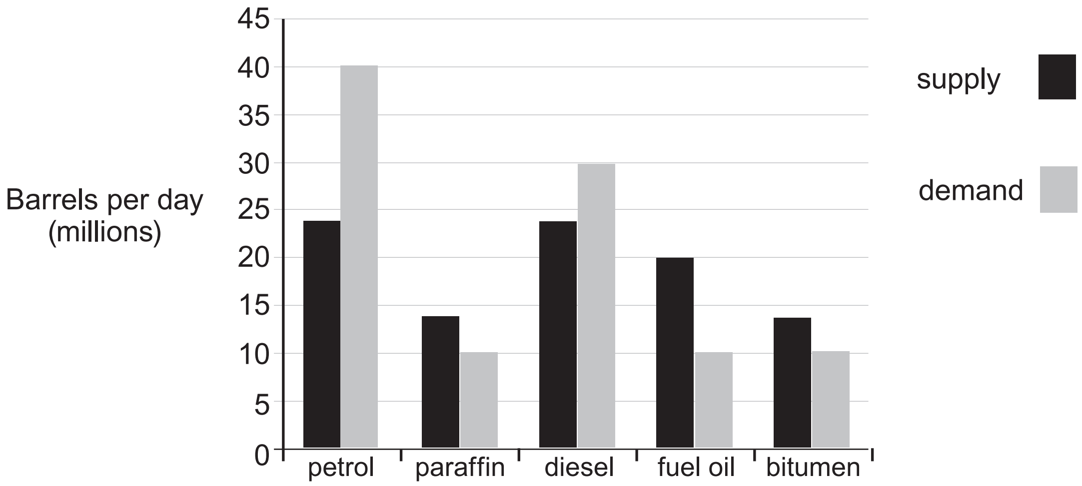
(b) The bar chart shows the **supply** and **demand** for some crude oil products.



Name the fractions for which the demand is greater than the supply. Suggest a reason why these fractions are in high demand. [2]

Fractions .... and ....

Reason ....

....

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(c) Plastic carrier bags are made from polythene. Each plastic carrier bag can take 500-1000 years to decompose and may never break down in landfill. Paper bags are not necessarily an environmentally friendly alternative. Manufacturing paper bags wastes a lot of natural resources. Even starch-based biodegradable bags use natural resources during their manufacture.

Supermarkets give customers a choice of buying single-use or re-usable polythene carrier bags.

Table 2 shows the number of both types of plastic bag sold in UK supermarkets from 2011 to 2013.

Year	2011	2012	2013
	Number of bags (millions)		
Single-use bags	7977	8079	8455
Re-usable bags	415	408	445

Table 2

(i) State **one** environmental problem related to the disposal of **all** types of carrier bag. [1]

....

(ii) Calculate the percentage of plastic bags sold in 2013 that were single-use bags. [2]

Percentage = ... %

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(iii) Although more plastic carrier bags were sold in 2013 than 2012, the total **mass** of those bags changed from 70 400 tonnes to 67 300 tonnes.

Put a tick (✓) in **two** boxes next to statements which could explain the reason for the change in mass. [2]

the bags were made the same thickness but from a less dense plastic	☐
customers re-used their plastic bags more often	☐
the bags were made from the same plastic but were thicker	☐
the bags were made from the same plastic but were thinner	☐
the bags were made the same thickness but from a more dense plastic	☐

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